

Databases & Data Management — Quick Reference

Backend Development — Resnet Academy

Core SQL Clauses

Clause	Purpose
SELECT ... FROM	Choose columns and source table
WHERE	Filter rows
JOIN	Combine rows from related tables
ORDER BY	Sort results
GROUP BY	Aggregate rows into groups

Common Migration Column Types

Method	SQL type
<code>\$table->id()</code>	BIGINT UNSIGNED, primary key
<code>\$table->string('title', 200)</code>	VARCHAR(200)
<code>\$table->text('description')</code>	TEXT
<code>\$table->foreignId('course_id')</code>	BIGINT UNSIGNED + FK
<code>\$table->timestamps()</code>	created_at / updated_at

Eloquent Relationship Cheat Sheet

Relationship	Example
hasMany	Course !' Modules
belongsTo	Module !' Course
belongsToMany	Module !" Group (pivot table)
hasOne	Resource !' ResourceDocument

Artisan Database Commands

Command	Purpose
<code>php artisan migrate</code>	Apply migrations
<code>php artisan migrate:fresh --seed</code>	Reset schema and reseed
<code>php artisan db:seed</code>	Run seeders only
<code>php artisan tinker</code>	Query the database interactively